Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 1 of 13 · Question 5 part b · 3 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Number			
5 (a)	any three from • rapid response (to a stimulus) (1) • involuntary/automatic (1) • no brain involvement (1) • protective (1)		3 max
(b)	• A = motor (1) • B = relay/association/connector (1) • C = sensory (1)		3
(c) (i)	• fewer branches/dendrons/dendrites (1) • shorter branches/ dendrons/dendrites (1) • fewer knobs/synapses (1)	R axons	3
(ii)	• transmits impulse (1) • across synapse/(synaptic) cleft (1) • stimulates impulse in next neurone (1)		3
(iii)	• secreted by synapses/synaptic knob (1) • fewer synapses/knobs (1) • less secreted by remaining synapses/knobs (1)		3

Total 15 marks

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 1 of 13 · Question 5 part b · 3 marks

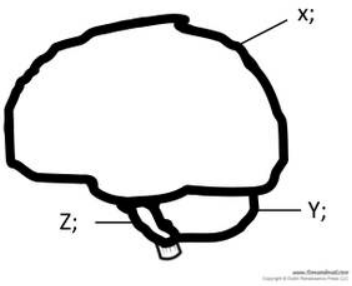
2020 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 2 of 13 · Question 7 part b.i · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

7 (a)		Letters can be anywhere within outline shown. Allow labels in place of than letters.	3
(b) (i)	Motor;		1
(ii)	Any one from the following: (sensory neurone): longer dendrites/shorter axon/cell body outside of spinal cord/CNS/in middle of neurone/transmit impulse to the CNS/brain/spinal cord (motor neurone) shorter dendrites/longer axon/cell body at one end of neurone/inside spinal cord/transmit impulses from CNS/brain/spinal cord	Answer must be comparative and state which type of neurone is being referred to	1
(iii)	• chemicals/neurotransmitters; • diffuse; • across a synapse; • bind to receptors (on post-synaptic membrane);		3
(c)	• stem cells are undifferentiated/unspecialised/can differentiate; • could be made to form nerve cells; • to replace damaged cells in brain;		1 1 1

Total question 7 = 11 marks

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 2 of 13 · Question 7 part b.i · 1 mark

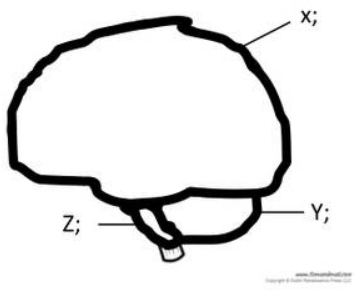
2019 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 3 of 13 · Question 7 part b.ii · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

7 (a)		Letters can be anywhere within outline shown. Allow labels in place of than letters.	3
(b) (i)	Motor;		1
(ii)	Any one from the following: (sensory neurone): longer dendrites/shorter axon/cell body outside of spinal cord/CNS/in middle of neurone/transmit impulse to the CNS/brain/spinal cord (motor neurone) shorter dendrites/longer axon/cell body at one end of neurone/inside spinal cord/transmit impulses from CNS/brain/spinal cord	Answer must be comparative and state which type of neurone is being referred to	1
(iii)	• chemicals/neurotransmitters; • diffuse; • across a synapse; • bind to receptors (on post-synaptic membrane);		3
(c)	• stem cells are undifferentiated/unspecialised/can differentiate; • could be made to form nerve cells; • to replace damaged cells in brain;		1 1 1

Total question 7 = 11 marks

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 3 of 13 · Question 7 part b.ii · 1 mark

2019 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 4 of 13 · Question 3 part b · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

number			
3 (a)	C hypothalamus; A does not control sweating B does not control sweating D does not control sweating		1
(b)	B motor; A transmits impulse from sensory to motor C alternative name for A with same function D transmits impulse to brain		1
(c)	• (sweat) evaporates from (the surface of) the skin; • using heat from body/blood; • cools body/reduces body temperature;		3
(d)	• runs in families; • must be faulty/mutated gene/chromosome/allele; • which transfers/passes on information to offspring;		3
(e)	obesity / high sugar intake / high BMI;		1
(f)	• exercise/activity of individual varies; • greater rate of aerobic respiration; • heat given off as by-product/increased body temperature; • external temperature varies; • hotter more sweating/cooler less sweating; • mass of individual varies; • volume of water in the body/fluid intake varies;	Allow reverse argument for mps 2 and 3	5 Total 14

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 4 of 13 · Question 3 part b · 1 mark

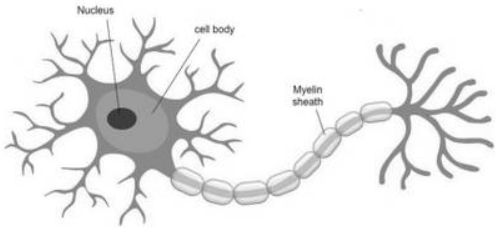
2021 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 5 of 13 · Question 7 part a.ii · 1 mark

2024 JUN · Paper 01 (regional R) · find part a.ii in this mark scheme extract

7 (a) (i)	receptor (cell)		(1)
(ii)	sensory neurone		(1)
(b) (i)	a cell that has particular features/adaptations/structures; to carry out a certain/specific function in the body;		(2)
(ii)	diagram; to include cell body at one end, with dendrites and an elongated axon twice the length of the cell body and axon ending; nucleus; cell body; myelin sheath;  	Label marks can be awarded if they draw a sensory or relay neurone	(4)
(iii)	transmits electrical impulses from the brain/CNS/spinal cord/ relay neurone; to effecter organs/muscle/gland;		(2)
(c)	100/4.0 x 10^{6}; = 0.000025 (s); = 0.025 (ms); = 2.5 x 10^{-2};	ecf for mps 2,3 and 4 Allow full marks for correct final answer	(4)

Total for Question 7 = 14 marks

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 6 of 13 · Question 1 part a.i · 1 mark

2024 JUN · Paper 01 · find part a.i in this mark scheme extract

1 (a)	(i)	sensory (neurone/cell);		(1)
	(ii)	axon;		(1)
(b)	(i)	Part R sweat gland; Part S blood vessels/capillaries/arterioles;		(2)
	(ii)	Any two from: • produces sweat/perspiration; • sweat evaporates (from skins surface); • using heat energy (from the blood);		(2)
	(iii)	• provides an insulating layer insulation; • traps heat (energy)/to prevent heat loss;	reject keeps body warm	(2)
(c)	(i)	• water is too hot/burn hands;		(1)
	(ii)	• use a thermometer; • to test the temperature of the water/to make sure the water is not too hot;		(2)
	(iii)	• (temperature) receptor;		(1)

Total for Question 1 = 12 Marks

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 7 of 13 · Question 3 part b.i · 3 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

3	(a)		• rapid/fast/eq. ; • involuntary/automatic action; • does not involve conscious thought; • protective/prevents (further) damage;	allow instant-mp1 ignore ref to brain ignore sub-conscious ignore unconscious	3
	(b)	(i)	• sensory; • relay/intermediate; • motor;	correct positions on diagram	3
		(ii)	correct direction from sensory to motor;	Mark correctly if there are more than 1 arrow, unle/wrongss they are in opposite direction	1
		(iii)	synapse/synaptic cleft;		1
					Total = 8

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 7 of 13 · Question 3 part b.i · 3 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 8 of 13 · Question 3 part a · 4 marks

2022 JUN · Paper 02 · find part a in this mark scheme extract

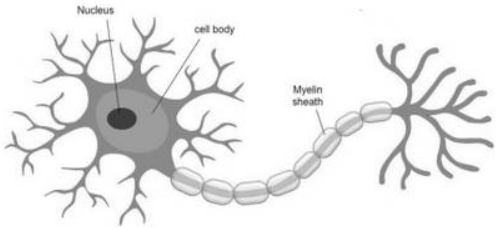
3 (a)	In the following order: • receptors; • sensory; • relay; • motor;		4
(b) (i)	synapse;		1
(ii)	• chemicals/neurotransmitters; • diffuse across synaptic clef/gap;		2
(c)	• 60 × 5; • 300 x 120 = 36 000;		2
			Total 9

Question number	Answer	Notes	Marks

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 9 of 13 · Question 7 part b.iii · 2 marks

2024 JUN · Paper 01 (regional R) · find part b.iii in this mark scheme extract

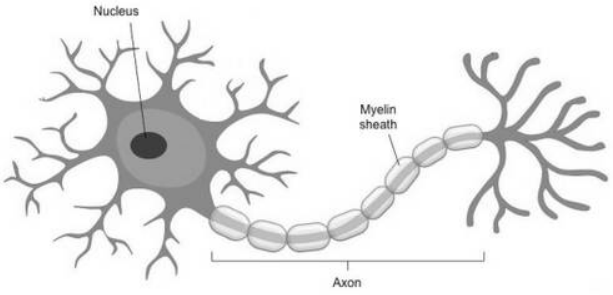
7 (a) (i)	receptor (cell)		(1)
(ii)	sensory neurone		(1)
(b) (i)	a cell that has particular features/adaptations/structures; to carry out a certain/specific function in the body;		(2)
(ii)	diagram; to include cell body at one end, with dendrites and an elongated axon twice the length of the cell body and axon ending; nucleus; cell body; myelin sheath;  	Label marks can be awarded if they draw a sensory or relay neurone	(4)
(iii)	transmits electrical impulses from the brain/CNS/spinal cord/ relay neurone; to effecter organs/muscle/gland;		(2)
(c)	100/4.0 x 10^{6}; = 0.000025 (s); = 0.025 (ms); = 2.5 x 10^{-2};	ecf for mps 2,3 and 4 Allow full marks for correct final answer	(4)

Total for Question 7 = 14 marks

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 10 of 13 · Question 3 part a.ii · 2 marks

2025 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

3	(a)	(i)		1 mark for diagram 3 marks for labels	4
		(ii)	• transmits electrical/nerve impulses; • from CNS/brain/spinal cord/relay neurone, to effector organs/named effector organ;		2
	(b)	(i)	Any six from 1 (student X) holds up the ruler between the thumb and forefinger (of other student); 2 then drops/releases the ruler; 3 (other) student catches the ruler; 4 record the reading on the ruler/distance fallen when caught; 5 repeat for <u>this</u> student; 6 using the same distance between thumb and forefinger; 7 find mean/average value of reading on ruler; 8 calculate reaction time (using formula); 9 repeat with <u>other</u> students;		6
		(ii)	the reading on the ruler/ reaction time;		1
		(iii)	independent variable not continuous/is discrete;		1
		(iv)	reaction time = √(0.15 ÷ 5); reaction time = 0.17 (s);	accept 0.173(2)	2
				Total for Question 3 = 16 marks	

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 11 of 13 · Question 5 part a · 3 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Number			
5 (a)	any three from • rapid response (to a stimulus) (1) • involuntary/automatic (1) • no brain involvement (1) • protective (1)		3 max
(b)	• A = motor (1) • B = relay/association/connector (1) • C = sensory (1)		3
(c) (i)	• fewer branches/dendrons/dendrites (1) • shorter branches/ dendrons/dendrites (1) • fewer knobs/synapses (1)	R axons	3
(ii)	• transmits impulse (1) • across synapse/(synaptic) cleft (1) • stimulates impulse in next neurone (1)		3
(iii)	• secreted by synapses/synaptic knob (1) • fewer synapses/knobs (1) • less secreted by remaining synapses/knobs (1)		3

Total 15 marks

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 11 of 13 · Question 5 part a · 3 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 12 of 13 · Question 8 part a · 3 marks

2021 JUN · Paper 01 · find part a in this mark scheme extract

number				
8	(a)	any three from • rapid response; • automatic; • to protect body; • from harmful effects/named example;		3
	(b)	correct position;		1
	(c)	any six from • impulse arrives at end of axon; • synapse; • causes release of neurotransmitter; • e.g. acetylcholine; • from vesicles; • diffuses across gap/cleft; • combines with/stimulates (post synaptic) receptors; • triggers impulse;		6

Total 10 marks

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 13 of 13 · Question 3 part a · 3 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

3	(a)		• rapid/fast/eq. ; • involuntary/automatic action; • does not involve conscious thought; • protective/prevents (further) damage;	allow instant-mp1 ignore ref to brain ignore sub-conscious ignore unconscious	3
	(b)	(i)	• sensory; • relay/intermediate; • motor;	correct positions on diagram	3
		(ii)	correct direction from sensory to motor;	Mark correctly if there are more than 1 arrow, unle/wrongss they are in opposite direction	1
		(iii)	synapse/synaptic cleft;		1
					Total = 8

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 13 of 13 · Question 3 part a · 3 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks